

Thesis Proposal

**Forecasting the USD–NGN Exchange Rate Using Information  
Theoretical Analysis, Machine Learning and xAI.**

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## 1.0 INTRODUCTION AND BACKGROUND

### 1.1 Introduction

Nigeria, as one of Africa's largest economies, occupies a strategic position in global financial markets. The Nigerian economy's heavy dependence on crude oil revenues, which account for approximately 90% of federal government earnings and 60% of government revenues, creates a unique vulnerability in its currency dynamics [1], making the Nigerian Naira (NGN) susceptible to external shocks from global oil market fluctuations as well as domestic economic policy adjustments.

The exchange rate serves as a barometer of Nigeria's economic health and stability. Fluctuations in this rate have implications for stakeholders, including importers who face increased costs during naira depreciation, exporters who benefit from competitive pricing, and policymakers who must balance exchange rate stability with economic growth objectives [2]. The volatility of this exchange rate has intensified recently due to several factors including oil price volatility, market reforms, political uncertainties, and economic crises.

Nigeria's exchange rate regime has changed significantly over time, moving from a fixed exchange rate system to various forms of managed float arrangements. The Central Bank of Nigeria (CBN) has implemented multiple exchange rate policies, including the adoption of a flexible regime in 2016, and the depegging of the naira in 2023, both which led to significant naira devaluation. These policy shifts, combined with pressures such as the oil price shock of 2014/16, the pandemic's economic impact, have created a complex and dynamic environment for exchange rate determination.

During these periods, traditional forecasting models have often failed to fully capture the underlying complexity and non-linear relationships in the Nigerian foreign exchange market. This has highlighted the need for more complex approaches that can adapt to sharp changes, and capture complex relationships among variables.

Recent advances in machine learning and artificial intelligence have opened new avenues for financial forecasting. Deep learning models, particularly Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU), have demonstrated solid performance in capturing temporal dependencies and non-linear patterns in financial time series data [3]. These models have shown particular promise in foreign exchange forecasting due to their ability to learn complex patterns and adapt to changing market conditions.

The Study in [4] showed that when fitting univariate ARIMA models to pre-COVID, COVID, and combined periods (Jan 2002–Jun 2022), the COVID window-only model actually forecasts Naira appreciation, directly contradicting the depreciation trend seen in the other periods and underscoring the inability of purely linear approaches to handle non-linear regime shifts and structural breaks.

However, the application of these advanced techniques to exchange rate forecasting in emerging markets, particularly in the Nigerian context, remains limited. Most existing studies focus on developed market currencies or apply single-method approaches without considering the specific structural characteristics of oil-dependent economies. Furthermore, the "black box" nature of deep learning models has raised concerns about interpretability and clarity, particularly in policy-making contexts where it is crucial to understand the rationale behind predictions.

### 1.2 Problem Statement

The existing literature on USD-NGN exchange rate forecasting reveals several critical and interconnected limitations that this research aims to address. Most studies employ single-method approaches, focusing either on traditional econometric models or machine learning techniques in isolation, which fails to leverage complementary methodological strengths and may miss important patterns [5]. Traditional econometric models such as ARIMA and EGARCH, while theoretically grounded and interpretable, often struggle to capture non-linear relationships and complex interdependencies that characterize modern foreign exchange markets [6]. Conversely, machine learning models demonstrate superior pattern recognition capabilities but lack the interpretability required for policy decision-making, creating a tension between accuracy and transparency [8].

A significant gap exists in the integration of causality-aware metrics in exchange rate forecasting models. Most studies treat input variables as equally important or rely on correlation-based feature selection methods, which may not capture the true directional relationships and information flows among variables—particularly problematic in the Nigerian context where relationships between oil prices, monetary policy variables, and exchange rates are complex and time-varying [7]. The absence of information flow analysis means models may include irrelevant features or miss important predictive relationships, leading to suboptimal forecasting performance.

Furthermore, while there has been substantial research on exchange rate forecasting in developed economies, studies specifically focused on emerging markets, particularly Nigeria, remain limited. The unique characteristics of Nigeria's oil-dependent economy, including structural breaks due to policy changes, oil price volatility, and external shocks, require specialized modeling approaches that account for these specific conditions [9]. Existing studies often apply models developed for stable, developed economies to the Nigerian context without adequate consideration of local economic dynamics, which may lead to models that perform poorly during periods of economic stress or policy changes, limiting their practical utility for Nigerian stakeholders.

These interconnected limitations, methodological isolation, lack of causal analysis, interpretability challenges, and limited context-specific research, collectively underscore the need for an integrated approach that combines information flow analysis with explainable machine learning models to improve both accuracy and interpretability of USD-NGN exchange rate forecasts.

### 1.3 Research Questions

Based on the identified limitations and gaps in existing literature, this research addresses the following key questions:

Primary Research Question: *How can the integration of information flow analysis with explainable machine learning models improve the accuracy and interpretability of USD-NGN exchange rate forecasts in Nigeria's oil-dependent economy?*

### 1.4 Aim and Objectives

#### 1.4.1 Research Aim

The primary aim of this research is to develop and evaluate a comprehensive hybrid forecasting framework that combines information flow analysis with explainable machine learning models to predict USD-NGN exchange rate movements. This framework will address the identified limitations in current approaches by integrating causal analysis, advanced machine learning techniques, and explainability methods to provide accurate, interpretable, and actionable exchange rate forecasts tailored to Nigeria's economic context.

### 1.4.2 Research Objectives

To achieve the stated aim, this research will pursue the following specific objectives:

**1: Data Collection and Preprocessing** Systematically collect and preprocess daily time-series data for USD-NGN exchange rates, Brent crude oil prices, Consumer Price Index (CPI) inflation rates, and CBN interest rates spanning the period from January 1995 to December 2024. This objective includes implementing robust data cleaning procedures, handling missing values, and performing necessary transformations to ensure data quality and consistency.

**Objective 2: Information Flow Analysis Implementation** Compute and analyze information flow metrics, specifically transfer entropy and mutual information, among the collected economic variables to identify directional dependencies and causal relationships. This analysis will reveal which variables provide the most valuable information for exchange rate prediction and inform feature selection processes.

**Objective 3: Hybrid-Model Implementation and Development** Develop and combine multiple forecasting models including traditional econometric approaches (ARIMA, ARIMAX) and deep learning models (LSTM, GRU), into an innovative hybrid/ensemble approach that combine the strengths of different methodologies. Each model will be carefully tuned and optimized for the specific characteristics of USD-NGN exchange rate data while integrating explainable AI techniques to the framework to provide transparency and lucid insights.

**Objective 4: Comprehensive Model Evaluation** Conduct rigorous evaluation of all developed models using multiple performance metrics including Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and directional accuracy measures. Statistical significance of performance differences will be assessed using the Diebold-Mariano test to ensure robust model comparison [10].

## 1.5. Scope of Study

### 1.5.1 Temporal Scope

This research will focus on a comprehensive analysis period spanning from January 1, 1995, to July 1st, 2025. This 30-year timeframe encompasses several critical economic periods in Nigeria's recent history, including:

- Return to democracy and removal of fixed exchange rate (1998-2000)
- The oil price boom period (2010-2014)
- The oil price crash and subsequent economic recession (2014-2016)
- Exchange rate regime transitions and policy reforms (2016-2020)
- The COVID-19 pandemic impact period (2020-2022)
- Post-pandemic recovery and current economic conditions (2022-2025)

This coverage ensures that the model is trained and tested across various economic cycles and policy regimes, enhancing their robustness and practical applicability.

### 1.5.2 Data Frequency and Variables

The study will utilize daily frequency data to capture short-term dynamics and provide forecasts. The core variables to be included are:

Primary Variable:

- USD-NGN exchange rate (dependent variable)

Explanatory Variables:

- Brent crude oil prices (primary commodity export)
- Consumer Price Index (CPI) inflation rate (monetary policy indicator)
- CBN policy interest rate (monetary policy tool).

Additional Control Variables (where possible):

- Nigerian foreign exchange reserves and Oil production volumes

### 1.5.3 Methodological Scope

The research will implement an ensemble forecasting approach utilizing a mix of:

Traditional Econometric Models:

- Autoregressive Integrated Moving Average (ARIMA)

Machine Learning Models:

- Long Short-Term Memory Networks (LSTM)

Information Flow Analysis:

- Transfer entropy computation
- Mutual information analysis

### 1.5.4 Geographical and Economic Context

The study is specifically focused on Nigeria's economic context and the USD-NGN exchange rate dynamics. While comparative references to other emerging markets may be included for benchmarking purposes, the primary analysis will center on Nigeria's unique economic characteristics, policy environment, and structural factors that influence exchange rate movements.

### 1.5.5 Study Limitations and Exclusions

- High-frequency intraday data will be excluded to maintain focus on daily and longer-term patterns (1-day, 5-day, 10-day)
- The study will focus on readily available, high-quality macroeconomic indicators
- Complex ensemble methods requiring extensive computational resources will be limited
- Real-time forecasting with live data streams is not within the current scope

## 2.0 LITERATURE REVIEW

### 2.1 Introduction

Exchange rate forecasting has been a central concern in international finance for decades. The efficient market hypothesis suggests that exchange rates should follow a random walk, making prediction difficult. However, empirical evidence has consistently shown that exchange rates exhibit predictable patterns, particularly at short-to-medium term horizons [20].

The literature distinguishes between fundamental and technical approaches to exchange rate forecasting. Fundamental analysis relies on macroeconomic variables such as interest rate differentials, inflation rates,

current account balances, and political stability indicators. Technical analysis, on the other hand, focuses on price patterns and trading volume data. Recent research has increasingly explored the integration of these approaches with advanced computational methods [21].

## 2.2 Traditional Econometric Approaches

### ARIMA Models and Extensions

Autoregressive Integrated Moving Average (ARIMA) models have been extensively applied to exchange rate forecasting due to their theoretical foundation and interpretability. The authors in [3] explored Nigerian exchange rate dynamics using ARIMA models and found reasonable forecasting performance for short-term horizons. However, their study highlighted limitations in capturing volatility clustering and non-linear patterns common in exchange rate data.

The extension of ARIMA models to include exogenous variables (ARIMAX) has shown promise in incorporating fundamental economic factors. In [9], it was demonstrated that ARIMAX models incorporating oil price variables achieved superior out-of-sample forecasting performance for Nigerian inflation rates, suggesting potential applications for exchange rate forecasting in oil-dependent economies.

### Volatility Modeling Approaches

GARCH-family models have been widely employed to capture the time-varying volatility characteristics of exchange rates. In [6] LSTM and EGARCH models were compared for Nigeria exchange rate volatility forecasting, finding that while EGARCH models effectively captured volatility clustering, they struggled with structural breaks and regime changes common in emerging market currencies.

The literature suggests that volatility models perform well during stable periods but may fail during crisis periods or policy regime changes. This limitation is particularly relevant for Nigeria, which has experienced multiple exchange rate regime transitions over the study period.

## 2.3 Machine Learning and Deep Learning Applications

### Long Short-Term Memory Networks

LSTM networks have gained significant attention in financial forecasting due to their ability to capture long-term dependencies in sequential data. Olanrewaju et al. [5] applied LSTM and GRU models to USD/NGN forecasting and demonstrated superior performance compared to traditional methods. Their study showed that deep learning models could effectively capture complex temporal patterns in exchange rate data.

However, most existing applications of LSTM to Nigerian exchange rate forecasting have focused primarily on prediction accuracy without adequate attention to model interpretability or economic intuition. This limitation reduces their practical utility for policy-making applications.

### Comparative Studies

In [14], a comprehensive comparison of ARIMA and LSTM models for foreign exchange forecasting was carried out across multiple currency pairs. The findings suggest that LSTM models generally outperform ARIMA for longer forecasting horizons, while ARIMA maintains competitiveness for short-term predictions. Importantly, they noted that performance varies significantly across different economic conditions and currency characteristics.



## 2.4 Hybrid and Ensemble Approaches

### ARIMA-Neural Network Hybrids

Hybrid approaches that combine the linear modeling capabilities of ARIMA with the non-linear pattern recognition abilities of neural networks have shown promising results. In [22], the authors developed a theoretical framework for ARIMA-ANN hybrid models, demonstrating that such combinations could outperform individual models by capturing both linear and non-linear components of time series data.

The authors in [24] specifically applied ARIMA-ANN hybrid models to exchange rate forecasting and found significant improvements in forecasting accuracy. Their approach involved using ARIMA to model linear components and neural networks to capture residual non-linear patterns.

### Ensemble Methods

Ensemble approaches that combine predictions from multiple models have shown potential for improving forecasting robustness and accuracy. In [23], various ensemble techniques for time series forecasting found that properly weighted combinations could reduce both bias and variance in predictions.

However, the application of ensemble methods to exchange rate forecasting in emerging markets remains limited, and there is a need for systematic evaluation of different combination strategies in the Nigerian context.

## 2.5 Information Flow Analysis in Financial Markets

### Transfer Entropy Applications

Transfer entropy provides an approach to measuring information flow between time series. In financial markets, transfer entropy has been used to identify lead-lag relationships, market interconnectedness, and information spillovers. Research in [16] demonstrated the application of transfer entropy to time series forecasting, showing that information flow metrics could inform feature selection and model structure decisions.

The application of transfer entropy to exchange rate forecasting in emerging markets is relatively limited, representing a significant research gap. Understanding information flows between oil prices, macroeconomic indicators, and exchange rates could provide valuable insights for both forecasting and policy formulation in Nigeria.

### Mutual Information and Causal Analysis

Mutual information provides another approach to measuring statistical dependencies between variables. Unlike correlation measures, mutual information can capture non-linear relationships and has been successfully applied in financial market analysis. The combination of transfer entropy and mutual information can provide a comprehensive picture of information flows and causal relationships among economic variables.

## 2.6 Explainable AI in Financial Forecasting

### SHAP and LIME Applications

The growing adoption of complex machine learning models in finance has created a need for explainability techniques that can provide insights into model decision-making processes. SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) have emerged as leading approaches for model interpretation [17], [18].

The authors in [17] demonstrated the application of explainable AI techniques to exchange rate forecasting, showing that SHAP values could provide meaningful insights into feature importance and model behavior. Their work highlighted the potential for explainable AI to bridge the gap between model accuracy and interpretability in financial applications.

## 2.7 Nigerian and African Context Studies

### Nigeria-Specific Research

In [2], ARIMA modelling was applied to Naira-Rupee exchange rates, highlighting the importance of oil price variables and structural breaks in the Nigerian economy context. In [7], a comprehensive analysis of Naira/USD exchange rate dynamics was carried out using the Box-Jenkins methodology, identifying key patterns and forecasting challenges..

In [4], the impact of the COVID-19 pandemic on Nigerian exchange rate dynamics was examined, and it was discovered that traditional forecasting models struggled to capture the effects of unprecedented external shocks. This finding highlights the need for more robust and adaptive forecasting approaches.

### Comparative African Studies

Research on other African currencies provides valuable insights for the Nigerian context. The authors of [10] applied LSTM networks to Ghanaian Cedi exchange rate forecasting and achieved promising results. Their methodology and findings offer transferable insights for Nigerian applications.

In [11], multivariate GARCH models were developed for inflation and exchange rate forecasting in Ghana, demonstrating the importance of considering interdependencies among macroeconomic variables. Their approach could be adapted for Nigerian applications with appropriate modifications.

## 2.8 Research Gaps and Opportunities

### Integration Gap

The literature review reveals a significant gap in the integration of information flow analysis with machine learning approaches for exchange rate forecasting. While both areas have shown promise individually, their combination remains largely unexplored, particularly in emerging market contexts.

### Context-Specific Gap

While general machine learning techniques have been applied to exchange rate forecasting, there is limited research that specifically accounts for the unique characteristics of oil-dependent emerging market economies like Nigeria.

### Methodological Innovation Opportunity

The combination of transfer entropy analysis, hybrid machine learning models, and explainable AI techniques represents a novel methodological approach that could advance both theoretical understanding and practical applications in exchange rate forecasting.

### 3.0 METHODOLOGY

#### 3.1 Design Overview

This study will employ a systematic four-stage approach: (1) data preprocessing and alignment, (2) information flow analysis for feature selection and engineering, (3) hybrid model development, and (4) comprehensive statistical evaluation. The methodology integrates traditional econometric foundations with modern machine learning techniques, guided by information-theoretic principles.

#### 3.2 Data Collection and Preprocessing

##### 3.2.1 Data Sources and Variables

Primary Variables:

- USD-NGN exchange rate (daily): Source - CBN official rates
- Brent crude oil prices (daily): OPEC/World Bank data
- CBN policy interest rate (monthly) ; Source - CBN
- CPI inflation index (monthly) : Source - National Bureau of Statistics

Period: January 1, 1995 to July 1, 2025 (11,140 days/observations)

##### 3.2.2 Data Alignment and Preprocessing

Frequency Harmonization: Monthly variables (interest rates, CPI) are forward-filled to daily frequency, creating step functions that change only on announcement dates—reflecting real market information availability.

Missing Data Treatment:

- Forward-fill for weekend/holiday gaps in exchange rates
- Linear interpolation for isolated missing values (<3 consecutive days)
- Exclude periods with >5 consecutive missing observations

Feature Engineering:

- Log returns: for exchange rates and oil prices
- Moving averages: 5-day, 20-day, and 60-day windows
- Volatility proxies: 20-day rolling standard deviation of returns

#### 3.3 Information Flow Analysis

##### 3.3.1 Transfer Entropy Computation

Transfer entropy measures directed information flow from source variable X to target variable Y:

Implementation:

- Discretization: Use equal-frequency binning with 6 bins (validated via cross-entropy minimization)
- Significance Testing: Bootstrap with 1,000 resamples;  $\alpha = 0.05$
- Rolling Windows: 250-day windows with 20-day step size to capture regime changes

### 3.3.2 Mutual Information Analysis

Mutual information captures symmetric, non-linear dependencies:

Implementation: Scikit-learn's `mutual_info_regression` with 10 nearest neighbors estimator.

## 3.4 Model Development

### 3.4.1 Benchmark Models

ARIMA: Selected via AIC with maximum orders (5,2,5)

### 3.4.2 Machine Learning Models

Proposed LSTM Architecture:

- Input layer: 60-day sequences with weighted features
- Hidden layers: 2 LSTM layers (128, 64 units) with 0.2 dropout
- Dense layers: 32 neurons  $\rightarrow$  1 output (next-day return)
- Activation: tanh (LSTM), ReLU (dense), linear (output)
- Optimizer: Adam ( $\text{lr}=0.001$ ,  $\beta_1=0.9$ ,  $\beta_2=0.999$ )
- Loss: Mean Squared Error with L2 regularization ( $\lambda=0.01$ )

Training Procedure:

- Batch size: 32 sequences
- Early stopping: patience=20 epochs on validation loss
- Feature scaling: StandardScaler applied to all inputs

### 3.4.3 Hybrid ARIMA-LSTM Model

Stage 1: Linear Component Fit ARIMA(p,d,q) to extract residuals

Stage 2: Non-linear Component Train LSTM on residual series information from ARIMA with information-weighted features:

- Target: ARIMA residuals
- Features: Weighted oil returns, interest rate changes, lagged residuals
- Architecture: Same as standalone LSTM but with 32-unit hidden layers

Stage 3: Hybrid Model - Final prediction combines both components to produce the optimized model.

## 3.5 Explainable AI Integration

### 3.5.1 SHAP Implementation

To improve interpretability, SHAP (SHapley Additive exPlanations) will be used to analyze model predictions. Since LSTMs are not natively tree-based, a gradient boosting surrogate model will be trained on the LSTM's input–output pairs. SHAP values will then be computed to:

- Identify the most influential features across time (global importance).
- Highlight regime-specific drivers of exchange rate movements.
- Provide local explanations for unusual or extreme predictions.

The results will offer both global insights (feature importance rankings) and local insights (case-specific explanations), ensuring the models are transparent and useful for policy and financial decision-making.

### 3.6 Benchmarking and Evaluation Framework

To ensure the robustness of the results, the developed model will be benchmarked against both classical econometric and machine learning baselines. This approach allows for a comparison between existing methods and the developed models introduced in this study.

#### I. Benchmark Models

- Statistical baselines: ARIMA and GARCH/EGARCH models, which are widely used in financial time series forecasting and volatility analysis.
- Machine learning baselines: Random Forest, XGBoost, and standard recurrent neural network models (LSTM/GRU), which have demonstrated competitive performance in financial forecasting tasks.
- Proposed models: Hybrid ARIMA-LSTM and its extended variants incorporating information flow analysis and explainable AI mechanisms (e.g., SHAP)

#### II. Evaluation Metrics

Model performance will be assessed using a combination of forecasting accuracy and financial relevance indicators:

- Mean Absolute Error (MAE) – measures average prediction error magnitude.
- Mean Absolute Percentage Error (MAPE) – provides a scale-free percentage error measure.
- Directional Accuracy (Hit Ratio): percentage of instances where the model correctly predicts the direction of exchange rate movements (upward or downward). This is critical for financial decision-making applications.

#### III. Statistical Validation

To verify that improvements in forecasting performance are not due to chance, pairwise statistical tests will be employed:

- Diebold-Mariano test: for comparing predictive accuracy between competing models.

By combining traditional econometric models, machine learning baselines, and the proposed hybrid framework, and by assessing performance through multiple accuracy and directional metrics with statistical significance tests, the evaluation will provide a balanced, realistic, and defensible measure of forecasting effectiveness.

### 3.7 Implementation Timeline

#### Phase 1: Data Collection and Preprocessing (Months 1)

- Data collection from multiple sources.
- Quality assurance and cleaning procedures.
- Feature engineering and transformation.
- Exploratory data analysis.

#### Phase 2: Information Flow Analysis (Months 2)

- Transfer entropy computation.
- Mutual information analysis.
- Causal relationship identification.
- Feature ranking and selection.

#### Phase 3: Model Development (Months 3)

- Machine learning models development.
- Hybrid model construction.
- Hyperparameter optimization.

#### Phase 4: Evaluation and Comparison (Months 4)

- Comprehensive model evaluation.
- Performance comparison and ranking.
- Robustness analysis.

#### Phase 5: Documentation and Dissemination (Months 5)

- Thesis writing and documentation.
- Presentation and defense preparation.

### 3.8 Tools and Environment

To implement the proposed hybrid forecasting framework, the following tools, libraries, and platforms will be employed:

1. Data Acquisition and Management
  - a. Python ( $\geq 3.8$ ): Core scripting language for data handling and modeling.
  - b. pandas: Loading, cleaning, and transforming time-series data (rolling windows, feature creation).
  - c. APIs and Data Feeds:
    - i. CBN API: USD–NGN daily rates and policy rates.
    - ii. National Bureau of Statistics (NBS) API: CPI inflation series.
2. Preprocessing and Feature Engineering
  - a. numpy: Array operations and numerical computations.
  - b. scikit-learn:
3. Econometric Modeling
  - a. statsmodels:
    - i. statsmodels.tsa.arima.model.ARIMA for ARIMA/ARIMAX.

4. Information Flow Analysis
  - a. pyinform: Computation of discrete transfer entropy metrics.
  - b. scikit-learn
    - i. Mutual information estimation
5. Deep Learning Components
  - a. PyTorch (alternative): LSTM/GRU modules and TorchData pipelines.
6. Explainable AI and Interpretation
  - a. SHAP:
    - i. shap.Explainer for global and local feature contributions.
7. Evaluation and Visualization
  - a. scikit-learn.metrics: RMSE, MAE, MAPE, and directional accuracy computations.
  - b. statsmodels.stats.diagnostic: Diebold–Mariano test implementation.
  - c. matplotlib: Plotting time-series forecasts, error distributions, and information-flow heatmaps.
8. Environment and version control
  - i. conda or pip-tools: Environment dependency management and version pinning.
  - ii. Git/GitHub: Version control for code, documentation, and experiment logs.
  - iii.
9. Hardware and Compute Resources
  - i. Local workstation: 16 GB RAM, multi-core CPU for initial development and testing.
  - ii. Microsoft Azure Compute

## 4.0 EXPECTED RESULTS AND DELIVERABLES

### 4.1 Forecasting Performance Improvements

Based on the comprehensive literature review and the proposed methodological innovations, several key outcomes are anticipated from this research:

- MAE Improvements: Mean Absolute Error reductions relative to baseline models.
- Directional Accuracy Enhancement: Improvement in directional accuracy (correctly predicting whether the exchange rate will increase or decrease)

### 4.2 Information Flow Analysis Insights

Key Driver Identification: The transfer entropy analysis is expected to reveal:

- The primary information source for USD-NGN movements, consistent with Nigeria's oil-dependent economy.
- How CBN Policy Interest Rates transmit information from monetary policy to exchange rates.
- Inflation Differentials demonstrating moderate but consistent information flow, particularly during periods of high inflation volatility.

Temporal Dynamics: The time-varying transfer entropy analysis is expected to identify:

- Regime changes during major economic events (2014-2016 oil crisis, 2020 COVID-19 pandemic)
- Strengthening oil-exchange rate linkages during periods of high oil price volatility

- Weakening traditional relationships during policy intervention periods

#### 4.3 Explainable AI Outcomes

The integration of Explainable AI is anticipated to provide:

Local Explanation Insights:

- Identification of specific instances where oil price movements dominate predictions
- Recognition of periods where monetary policy signals override oil price effects
- Detection of structural break periods where historical patterns lose predictive power
- Instance-specific feature contributions that vary significantly across different market conditions
- Identification of threshold effects where small changes in key variables lead to large prediction changes

#### 4.4 Economic and Policy Insights

Monetary Policy Implications: The research is expected to quantify:

- Policy Transmission Delays: Measurement of the time lag between CBN policy rate changes and their impact on exchange rate expectations (expected 2-5 business days)
- Policy Effectiveness Metrics: Quantification of how effectively interest rate changes influence exchange rate stability during different economic periods
- Optimal Intervention Timing: Identification of market conditions where policy interventions are most effective

Oil Market Dependency Quantification:

- Sensitivity Coefficients: Precise measurement of exchange rate response to oil price changes
- Asymmetric Effects: Documentation of whether positive and negative oil price shocks have different impacts on the Naira
- Threshold Effects: Identification of critical oil price levels that trigger regime changes in exchange rate behavior

#### 4.5 Model Robustness and Stability

Stress Testing Results: The models are expected to demonstrate:

- Crisis Period Performance: Maintained reasonable accuracy during the 2014-2016 oil crisis and 2020 pandemic periods, with performance degradation limited to 15-20%
- Regime Change Adaptation: Ability to adapt to policy regime changes within 10-20 trading days
- Volatility Resilience: Stable performance across different volatility regimes

#### 4.6 Practical Application Outcomes

- Risk Management Enhancement: Improved Value-at-Risk calculations with better accuracy
- Client Advisory Services: Interpretable forecasts that can be communicated effectively to corporate clients
- Cash Flow Planning: Enhanced ability to plan foreign currency cash flows with greater confidence



- **Cost Management:** Reduced foreign exchange transaction costs through better timing of currency conversions

#### 4.7 Academic Contributions

Methodological Innovations:

- Comprehensive application of transfer entropy to Nigerian exchange rate forecasting
- Novel hybrid architecture combining information flow metrics with explainable deep learning
- Systematic framework for integrating economic intuition with machine learning predictions

Theoretical Contributions:

- Quantification of information spillovers in oil-dependent emerging market currencies
- Documentation of regime-dependent relationships between macroeconomic variables and exchange rates

### 5.0 CONCLUSION

#### 5.1 Research Significance and Innovation

This proposed thesis attempts to present an approach to USD-NGN exchange rate forecasting that addresses critical gaps in existing literature while providing practical solutions for Nigerian economic stakeholders. The integration of information flow analysis with explainable machine learning attempts to enhance both forecasting accuracy and interpretability in emerging market currency prediction.

The research's novelty lies in its systematic combination of three key elements:

- (1) information flow metrics that reveal causal relationships and directional dependencies among economic variables.
- (2) hybrid machine learning models that leverage both traditional econometric foundations and modern deep learning capabilities.
- (3) explainable AI techniques that provide transparent insights into model decision-making processes. This integrated approach directly addresses the identified limitations in current forecasting frameworks while maintaining practical relevance for policy and business applications.

The anticipated results will enhance understanding of exchange rate dynamics in oil-dependent emerging economies, provide improved forecasting tools for stakeholders, and establish a framework for similar applications in other contexts. This research represents an important step toward bridging the gap between advanced analytical methods and practical economic applications in developing country contexts.

#### 5.2 Expected Impact.

**Theoretical Contributions:** The research will advance theoretical understanding of exchange rate dynamics in oil-dependent emerging economies by quantifying information spillovers and causal relationships among key macroeconomic variables. The application of transfer entropy and mutual information analysis to Nigerian exchange rate forecasting will provide new insights into the temporal dynamics of economic relationships and their implications for predictive modeling.

**Methodological Contributions:** The development of hybrid models that systematically integrate information flow analysis with machine learning techniques represents a significant methodological innovation. The proposed framework provides a replicable approach for combining economic intuition with advanced computational methods, offering a template for similar applications in other emerging market contexts.

**Industry Applications:** Financial institutions, import/export businesses, and other stakeholders will benefit from more accurate and interpretable exchange rate forecasts. The improved directional accuracy and reduced forecasting errors will enhance risk management practices, trading strategies, and business planning processes across the Nigerian financial sector..

### 5.3 Limitations and Future Research Directions

**Acknowledged Limitations:** While the proposed research addresses significant gaps in existing literature, several limitations should be acknowledged. The focus on daily frequency data may miss important high-frequency patterns that could enhance short-term forecasting accuracy. The exclusion of sentiment-based variables and alternative data sources represents a potential area for future enhancement.

**Future Research Opportunities:** The successful completion of this research will open several avenues for future investigation. Extensions to include alternative data sources such as social media sentiment, satellite-derived economic indicators, and cross-country information spillovers could further enhance forecasting performance. The methodology could be adapted for other emerging market currencies and extended to longer-term forecasting horizons.

## REFERENCES

- [1] P. O. Odion, M. M. Lawal, and A. Abdulrauf, "A comparative analysis of an enhanced hybrid model for predicting dollar against naira exchange rate using deep learning and statistical methods," *J. Comput. Theor. Appl.*, vol. X, no. X, pp. xx–xx, 2024, doi: 10.62411/jcta.12513.
- [2] A. C. Gabriel, "Modelling naira–rupee exchange rate: An ARIMA framework," *Asian J. Econ. Finance Manage.*, vol. 4, no. 1, pp. 416–425, 2022.
- [3] C. A. Awogbemi, A. I. Awoniyi, and O. E. Ajiboye, "An exploration of Nigerian exchange rate using ARIMA model," *Asian J. Probab. Stat.*, vol. 27, no. 1, pp. 69–80, 2025, doi: 10.9734/ajpas/2025/v27i1704.
- [4] E. Oduntan, "An ARIMA modeling of the exchange rate of the Nigerian naira to the US dollar with COVID-19 pandemic event in focus," *J. Stat. Model. Anal.*, vol. 6, no. 1, pp. xx–xx, 2024, doi: 10.22452/josma.vol6no1.1.
- [5] M. O. Olanrewaju, S. Luka, and F. O. Echobu, "Prediction of forex prices on USD/NGN using deep learning (LSTM and GRU) techniques," *J. Inf. Syst. Inform.*, vol. 5, no. 4, pp. 1609–1622, 2023, doi: 10.51519/journalisi.v5i4.606.
- [6] S. O. Adams and J. I. Uchema, "Nigeria exchange rate volatility: A comparative study of recurrent neural network LSTM and EGARCH models," *J. Artif. Intell. Big Data*, vol. 4, no. 2, pp. 61–73, Jun. 2024, doi: 10.31586/jaibd.2024.983.
- [7] T. Nyoni, "Modeling and forecasting naira/USD exchange rate in Nigeria: A Box–Jenkins ARIMA approach," *MPRA Paper* No. 88622, 2018. [Online]. Available: <https://mpra.ub.uni-muenchen.de/88622>. [Accessed: Jul. 23, 2025].
- [8] A. Ikoku, "Modeling and forecasting currency in circulation for liquidity management in Nigeria," *CBN J. Appl. Stat.*, vol. 5, no. 1, pp. xx–xx, 2021.
- [9] M. O. Adenomon and F. O. Madu, "Comparison of the out-of-sample forecast for inflation rates in Nigeria using ARIMA and ARIMAX models," in *Time Series Analysis – New Insights*, IntechOpen, 2022. doi: 10.5772/intechopen.XXXXX.
- [10] A. F. Adekoya, I. K. Nti, and B. A. Weyori, "Long short-term memory network for predicting exchange rate of the Ghanaian cedi," *FinTech*, vol. 1, no. 1, pp. 25–43, 2022, doi: 10.3390/fintech1010002.
- [11] E. N. N. Nortey *et al.*, "Modeling inflation rates and exchange rates in Ghana: Application of multivariate GARCH models," *SpringerPlus*, vol. 4, art. 86, 2015, doi: 10.1186/s40064-015-1082-7.
- [12] D. C. Yildırım *et al.*, "Forecasting directional movement of forex data using LSTM with technical and macroeconomic indicators," *Financial Innov.*, vol. 7, art. 1, 2021, doi: 10.1186/s40854-020-00220-2.
- [13] J. F. Pfahler, "Exchange rate forecasting with advanced machine learning methods," *J. Risk Financ. Manage.*, vol. 15, no. 1, art. 2, 2022, doi: 10.3390/jrfm15010002.

- [14] F. García *et al.*, “Foreign exchange forecasting models: ARIMA and LSTM comparison,” in *Proc. 9th Int. Conf. Time Series Forecasting, Eng. Proc.*, vol. 39, art. 81, 2023, doi: 10.3390/engproc2023039081.
- [15] S. Atha and B. K. Bolla, “Do deep learning models and news headlines outperform conventional prediction techniques on forex data?,” *arXiv preprint arXiv:2205.10743*, May 2022.
- [16] P. Papaioannou *et al.*, “Time series forecasting using manifold learning,” *arXiv preprint arXiv:2110.03625*, Oct. 2021.
- [17] S. Meng *et al.*, “Enhancing exchange rate forecasting with explainable deep learning models,” *arXiv preprint arXiv:2410.19241*, Oct. 2024.
- [18] H. Ding *et al.*, “EUR-USD exchange rate forecasting based on information fusion with large language models and deep learning methods,” *arXiv preprint arXiv:2408.13214*, Aug. 2024.
- [19] M. Y. Rahat *et al.*, “Advancing exchange rate forecasting: Leveraging machine learning and AI for enhanced accuracy in global financial markets,” *arXiv preprint arXiv:2506.09851*, Jun. 2025.
- [20] S. Ahmed *et al.*, “FLF-LSTM: A Novel Prediction System Using Forex Loss Function,” *Appl. Soft Comput.*, vol. 97, p. 106780, 2020, doi:10.1016/j.asoc.2020.106780.
- [21] A. Amo Baffour *et al.*, “A Hybrid Artificial Neural Network-GJR Modeling Approach to Forecasting Currency Exchange Rate Volatility,” *Neurocomputing*, vol. 365, pp. 285–301, 2019, doi:10.1016/j.neucom.2019.07.088.
- [22] M. Khashei and M. Bijari, “A New Class of Hybrid Models for Time Series Forecasting,” *Expert Systems Appl.*, vol. 39, no. 12, pp. 4344–4352, 2012, doi:10.1016/j.eswa.2012.02.063.
- [23] N. Zougagh *et al.*, “Artificial Intelligence Hybrid Models for Improving Forecasting Accuracy,” *Procedia Comput. Sci.*, vol. 184, pp. 817–822, 2021, doi:10.1016/j.procs.2021.03.104.
- [24] R. Mucaj and V. Sinaj, “Exchange rate forecasting using ARIMA, NAR and ARIMA–ANN hybrid model,” *J. Multidiscip. Eng. Sci. Technol.*, vol. 4, no. 3, pp. 8581–8586, 2017.
- [25] F. Pandurang and K. Kumar, “Ensemble computations on stock market: A standardized review for future directions,” in *Proc. IEEE Int. Conf. Electr., Comput. Commun. Technol. (ICECCT)*, 2019, pp. 1–6, doi: 10.1109/ICECCT.2019.8869215.